

Lecture 6 Exercise Solutions

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1 Important Equations

Confidence Interval for the Mean

The confidence interval (CI) for the mean of the random normally distributed variable $X \sim N(\mu, \sigma^2)$ with a confidence level of $1 - \alpha$ have two cases. First for known variance:

$$\left[\bar{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right] \quad (1)$$

If the variance of the population is estimated from the sample then:

$$\left[\bar{X} \pm t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right] \quad (2)$$

A one-sided CI can also be constructed for the mean in this case with known variance:

$$\left[\bar{X} - Z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty \right], \quad \left[-\infty, \bar{X} + Z_{\alpha} \frac{\sigma}{\sqrt{n}} \right] \quad (3)$$

Confidence Interval for the Variance

For a iid and normally distributed random variable $X \sim N(\mu, \sigma^2)$ the CI for the variance is given as:

$$\left[\frac{(n-1)s^2}{\chi_{n-1, 1-\frac{\alpha}{2}}}, \frac{(n-1)s^2}{\chi_{n-1, \frac{\alpha}{2}}} \right] \quad (4)$$

Approximate Confidence Interval on a Binomial Proportion

If $\hat{p}$ is the proportion of observations in a random sample of size n that belongs to a class of interest, an approximate $100(1 - \alpha)\%$ CI on the proportion p of the population that belongs to this class is:

$$\left[\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right] \quad (5)$$

An approximate one-sided interval can also be calculated as:

$$\left[\hat{p} - Z_{\alpha} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \infty \right], \quad \left[-\infty, \hat{p} + Z_{\alpha} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right] \quad (6)$$

2 Exercise 8.1.3

A random sample has been taken from a normal distribution and the following confidence intervals constructed using the same data: (38.02, 61.98) and (39.95, 60.05)

a. what is the value of the sample mean?

We know that the normal distribution is symmetric and therefore have the mean in the middle. This means we can get the sample mean by finding the middle of the interval:

$$\bar{X} = 38.02 + \frac{61.98 - 38.02}{2} = 50$$

Where we get the same mean from the other confidence interval since they come from the same data:

$$\bar{X} = 39.95 + \frac{60.05 - 39.9}{2} = 50$$

b. One of these intervals is a 95% CI and the other is a 90% CI. Which one is the 95% CI and why?

If we want a higher confidence level for the confidence interval then it will be wider. This means that the confidence interval (38.02, 61.98) is the 95%.

3 Exercise 8.1.4

A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that the gain is normally distributed with standard deviation $\sigma = 20$.

a. How large must n be if the length of the 95% CI is to be 40?

We want to find a value of n in eq. 1. To do this we first find using MATLAB:

$$Z_{\frac{\alpha}{2}} = \text{norminv}\left(1 - \frac{\alpha}{2}\right) = \text{norminv}\left(1 - \frac{0.05}{2}\right) = 1.96$$

We can now isolate for n since we know that the CI should have a length of 40. Because the mean is the middle of the interval we can find n from half the length of

the interval i.e. $40/2 = 20$:

$$20 = Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \Rightarrow n = \left(Z_{\frac{\alpha}{2}} \frac{\sigma}{20} \right)^2 = \left(1.96 \frac{20}{20} \right)^2 = 3.84 \approx 4$$

Since sample size is an integer then $n = 4$.

b. How large must n be if the length of the 99% CI is to be 40?

We again use MATLAB:

$$Z_{\frac{\alpha}{2}} = \text{norminv} \left(1 - \frac{\alpha}{2} \right) = \text{norminv} \left(1 - \frac{0.01}{2} \right) = 2.5758$$

We can now find sample size with the same equation as in a.:

$$n = \left(Z_{\frac{\alpha}{2}} \frac{\sigma}{20} \right)^2 = \left(2.5758 \frac{20}{20} \right)^2 = 6.63 \approx 7$$

4 Exercise 8.1.12

a. Find a 95% confidence interval for the mean adhesion.

From MATLAB with $\alpha = 0.05$ we find:

$$Z_{\frac{\alpha}{2}} = \text{norminv} \left(1 - \frac{\alpha}{2} \right) = \text{norminv} \left(1 - \frac{0.05}{2} \right) = 1.96$$

The mean is calculated from the five readings:

$$\bar{X} = \frac{1}{5}(2.69 + 5.76 + 2.67 + 1.62 + 4.12) = 3.37$$

The exercise states a standard deviation of $\sigma = 0.66$ meaning the CI is:

$$\left[\bar{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right] = \left[3.37 \pm 1.96 \cdot \frac{0.66}{\sqrt{5}} \right] = [2.79; 3.94]$$

b. If the scientists want the confidence interval to be no wider than 0.55 dyne-cm², how many observations should they take?

We use the same equation from exercise 8.1.4 to calculate n from half the length of the confidence interval:

$$\frac{0.55}{2} = Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{0.66}{\sqrt{n}} \Rightarrow n = 22$$

5 Exercise 8.2.1

A random sample has been taken from a normal distribution. Output from a software package follows:

Variable	N	Mean	SE Mean	StDev	Variance	Sum
x	10	?	0.507	1.605	?	251.848

a. Fill in the missing quantities.

The variance is calculated from the standard deviation:

$$\sigma^2 = 1.605^2 = 2.58$$

The mean is calculated from the sum and number of samples:

$$\bar{X} = \frac{251.848}{10} = 25.1848$$

Find a 95% CI on the population mean.

Because we now estimate the population variance using the sample we have to use the student t distribution. Using MATLAB we calculate:

$$t_{n-1, \frac{\alpha}{2}} = \text{tinv} \left(1 - \frac{\alpha}{2}, N - 1 \right) = \text{tinv} \left(1 - \frac{0.05}{2}, 10 - 1 \right) = 2.2622$$

From eq. 2 the CI is:

$$\left[\bar{X} \pm t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right] = \left[25.1848 \pm 2.2622 \cdot \frac{\sqrt{2.58}}{\sqrt{10}} \right] = [24.04; 26.33]$$

6 Exercise 8.2.9

The compressive strength of concrete is being tested by a civil engineer who tests 12 specimens and obtains the following data:

$$X = [2216, 2237, 2249, 2204, 2225, 2301, 2281, 2263, 2318, 2255, 2275, 2295];$$

a. Check the assumption that the compressive strength is normally

distributed. Include a graphical display in your answer.

To check if the data is normally distributed we can make a QQ-plot. From fig. 1 we see that the data follows the straight line only with small deviations at the ends.

This mean we can assume that the compressive strength is normally distributed.

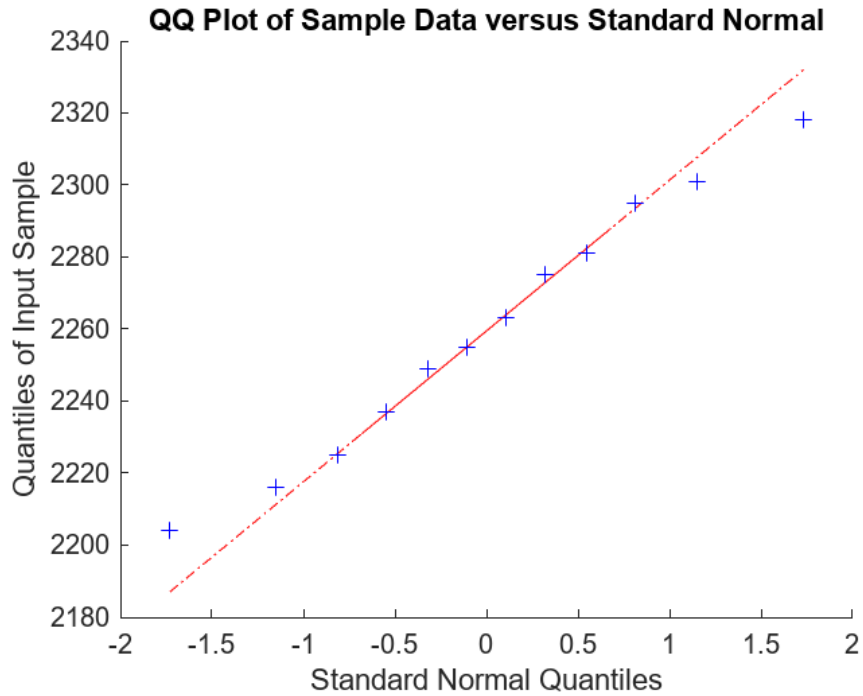


Figure 1: QQ-plot of the compressive strength data.

b. Construct a 95% two-sided confidence interval on the mean strength.

To find the confidence interval we use eq. 2 since we have to estimate the variance.

First we find the sample mean and variance:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 2259.9, \quad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2} = 35.693$$

Using MATLAB we find:

$$t_{n-1, \frac{\alpha}{2}} = \text{tinv} \left(1 - \frac{\alpha}{2}, n - 1 \right) = \text{tinv} \left(1 - \frac{0.05}{2}, 12 - 1 \right) = 2.201$$

The CI can now be calculated:

$$\left[\bar{X} \pm t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right] = \left[2259.9 \pm 2.201 \cdot \frac{35.693}{\sqrt{12}} \right] = [2237.3; 2282.5]$$

c. Construct a 95% lower confidence bound on the mean strength.

Compare this bound with the lower bound of the two-sided confidence interval and discuss why they are different.

To find the lower confidence bound we use eq. 3 but for unknown variance i.e. we have to use student t distribution and sample standard deviation. First with MATLAB:

$$t_{n-1,\alpha} = \text{tinv}(1 - 0.05, 12 - 1) = 1.7959$$

Now we have all other values from before meaning we can calculate the lower CI:

$$\left[\bar{X} - t_{n-1,\alpha} \frac{\sigma}{\sqrt{n}}, \infty \right] = \left[2259.9 - 1.7959 \frac{35.693}{\sqrt{12}}, \infty \right] = [2241.5, \infty[$$

7 Exercise 8.2.12

A healthcare provider monitors the number of CAT scans performed each month in each of its clinics. The most recent year of data for a particular clinic follows (the reported variable is the number of CAT scans each month expressed as the number of CAT scans per thousand members of the health plan). 2.31, 2.09, 2.36, 1.95, 2.25, 2.16, 2.07, 1.88, 1.94, 1.97 and 2.02.

a. Find a 95% two-sided CI on the mean μ number of CAT scans performed each month at this clinic.

Data in MATLAB format:

$$X = [2.31, 2.09, 2.36, 1.95, 1.98, 2.25, 2.16, 2.07, 1.88, 1.94, 1.97, 2.02];$$

First the sample mean is calculated:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 2.0817$$

And the sample standard deviation:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2} = 0.1564$$

Because we estimate the variance we use student t distribution:

$$t_{n-1, \frac{\alpha}{2}} = \text{tinv} \left(1 - \frac{\alpha}{2}, n - 1 \right) = \text{tinv} \left(1 - \frac{0.05}{2}, 12 - 1 \right) = 2.201$$

The two-sided CI can now be calculated:

$$\left[\bar{X} \pm t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right] = \left[2.0817 \pm 2.201 \cdot \frac{0.1564}{\sqrt{12}} \right] = [1.98; 2.18]$$

b. Historically, the mean number of scans performed by all clinics in the system has been 1.95. Is there any evidence that this particular clinic performs more CAT scans on average than the overall system average?

We can see from the two-sided interval that the lower bound is above the system average of 1.95. However we can construct a one-sided upper CI for the mean to verify. With MATLAB:

$$t_{n-1, \alpha} = \text{tinv}(1 - \alpha, n - 1) = \text{tinv}(1 - 0.05, 12 - 1) = 1.7959$$

Meaning the CI is:

$$\left[\bar{X} - t_{n-1, \alpha} \frac{s}{\sqrt{n}}, \infty \right] = \left[2.0817 - 1.7959 \frac{0.1564}{\sqrt{12}}, \infty \right] = [2, \infty[$$

Which supports the evidence that the clinic performs a higher number of CAT scans than the system average.

8 Exercise 8.3.3

MATLAB friendly data format:

`X = [23.3, 21.7, 21.6, 21.7, 21.3, 20.7, 20.9, 20.1];`

First we check the assumption of normality in the population using a normal probability plot. This is done in MATLAB using the command "normplot(X)".

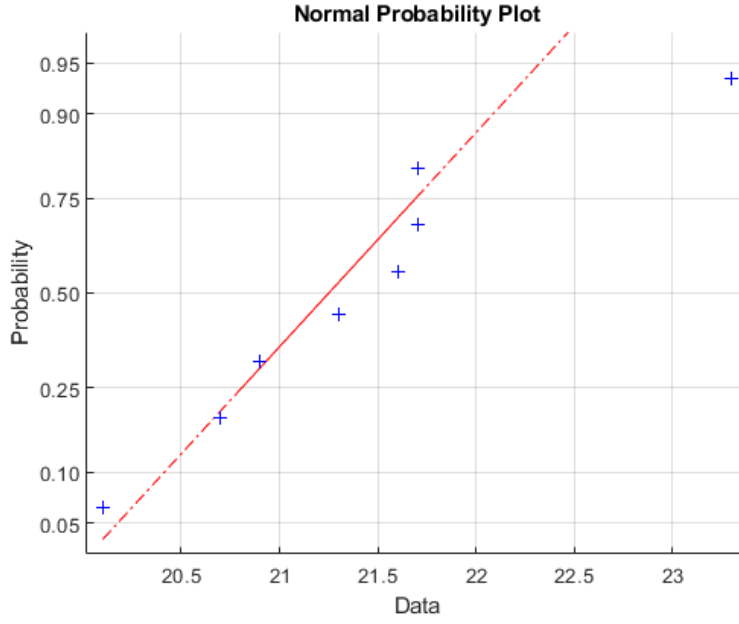


Figure 2: Normal probability plot of the data.

From the normal probability plot (fig. 2) we can see that the data mostly follows the line. This means the assumption of normality in the population is valid. We can now construct the 95% CI for the standard deviation using eq. 4. First we find the sample mean and variance:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 21.41, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = 0.8955$$

We now calculate using MATLAB:

$$\chi_{n-1, \frac{\alpha}{2}} = \text{chi2inv}\left(\frac{\alpha}{2}, n-1\right) = \text{chi2inv}\left(\frac{0.05}{2}, 8-1\right) = 1.6899$$

$$\chi_{n-1, 1-\frac{\alpha}{2}} = \text{chi2inv}\left(1 - \frac{\alpha}{2}, n-1\right) = \text{chi2inv}\left(1 - \frac{0.05}{2}, 8-1\right) = 16.013$$

Since eq. 4 is given for the variance we can get the CI for the standard deviation by taking the square root:

$$\left[\sqrt{\frac{(n-1)s^2}{\chi_{n-1, 1-\frac{\alpha}{2}}}}, \sqrt{\frac{(n-1)s^2}{\chi_{n-1, \frac{\alpha}{2}}}} \right] = \left[\sqrt{\frac{(8-1) \cdot 0.8955^2}{16.013}}, \sqrt{\frac{(8-1) \cdot 0.8955^2}{1.6899}} \right] = [0.6257; 1.926]$$

9 Exercise 8.4.1

The 2004 presidential election exit polls from the critical state of Ohio provided the following results. The exit polls had 2020 respondents, 768

of whom were college graduates. Of the college graduates, 412 voted for George Bush.

a. Calculate a 95% confidence interval for the proportion of college graduates in Ohio who voted for George Bush.

To find the CI we need to use eq. 5. The estimated proportion can be found as the number of college graduates who voted for George Bush divided by the total amount of college graduates:

$$\hat{p} = \frac{412}{768} = 0.5365$$

Using MATLAB we now calculate with $\alpha = 0.05$:

$$Z_{\frac{\alpha}{2}} = \text{norminv}\left(1 - \frac{0.05}{2}\right) = 1.96$$

Now the 95% CI for p which is the proportion of college graduates who voted for George Bush can be calculated:

$$\left[\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}\right] = \left[0.5365 \pm 1.96 \cdot \sqrt{\frac{0.5365 \cdot (1 - 0.5365)}{768}}\right] = [0.501; 0.572]$$

b. Calculate a 95% lower confidence bound for the proportion of college graduates in Ohio who voted for George Bush

We can find the lower bound using a one-sided CI given in eq. 6. We only need to calculate:

$$Z_{\alpha} = \text{norminv}(1 - \alpha) = \text{norminv}(1 - 0.05) = 1.6449$$

The lower bound 95% CI for p is then:

$$\left[\hat{p} - Z_{\alpha} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}, \infty\right] = \left[0.5365 - 1.6449 \cdot \sqrt{\frac{0.5365(1 - 0.5365)}{768}}, \infty\right] = [0.507, \infty[$$